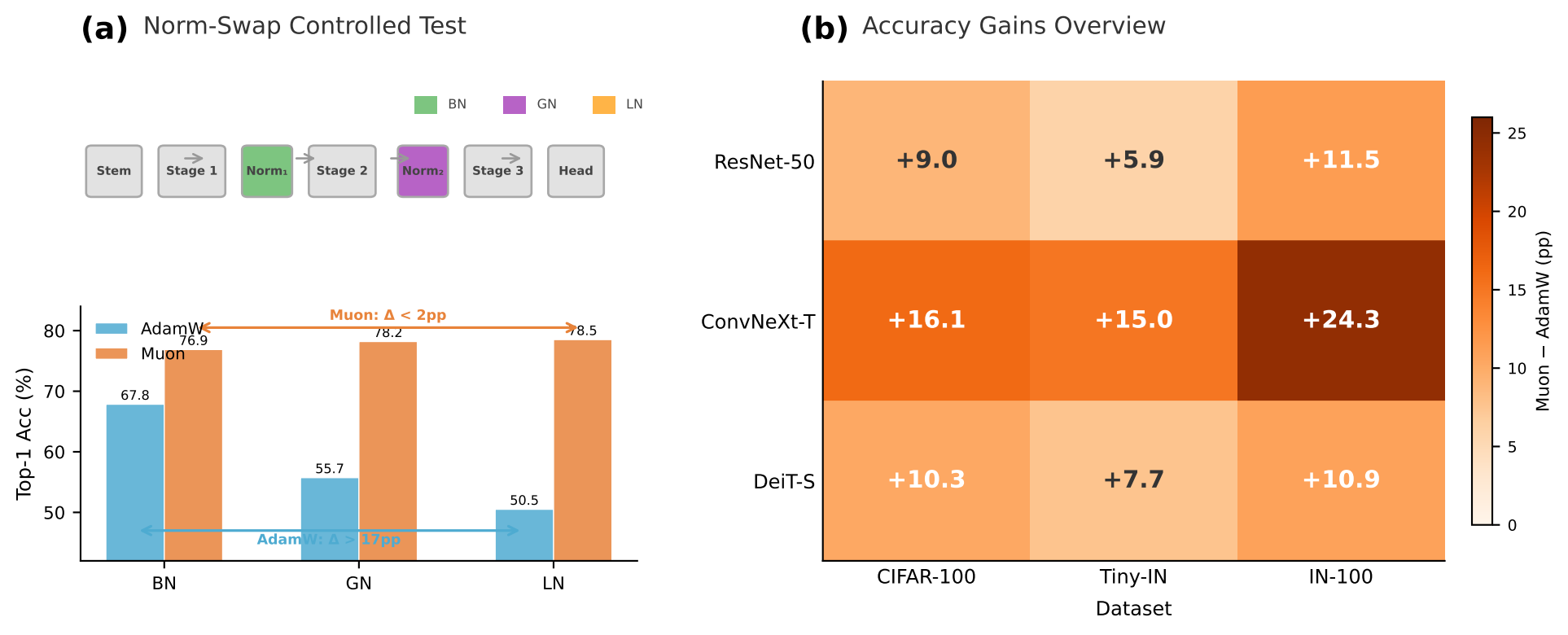
(a) Norm-Swap Controlled Test
BN
GN
LN
Stem
Stage 1
Norm1
Stage 2
Norm2
Stage 3
Head
AdamW
Muon
Muon: Δ < 2pp
AdamW: Δ > 17pp
Top-1 Acc (%)
80
70
60
50
76.9
67.8
78.2
55.7
78.5
50.5
BN
GN
LN
(b) Accuracy Gains Overview
ResNet-50
ConvNeXt-T
DeiT-S
+9.0
+5.9
+11.5
+16.1
+15.0
+24.3
+10.3
+7.7
+10.9
CIFAR-100
Tiny-IN
IN-100
Dataset
Muon − AdamW (pp)
25
20
15
10
5
0